Layer 1

Layer 2

Layer 3

Layer 4

Layer 6

Layer 6 (selected)

Layer 7

Layer 9

Layer 9 (selected)

Layer 10

Layer 11

Layer 12 (selected)

Layer 13

Layer 14

Layer 15 (selected)

Layer 16

Layer 17

Layer 19 (selected)

Layer 19

Layer 20

Layer 21

Layer 22

Layer 23

Layer 24

Multi-Layer Feature Extraction from WavLM

Layer Fusion

Mean

Concat

Weighted

Fused Features
1024-dim

Layer roles:
• Layer 1: Low-level acoustic
• Layer 6-12: Mid-level phonetic
• Layer 19-24: High-level semantic
Fusion combines multiple perspectives